Personalization in Goal-oriented Dialog

Chaitanya K. Joshi, Fei Mi, and Boi Faltings
Artificial Intelligence Laboratory
École Polytechnique Fédérale de Lausanne (EPFL)
Lausanne, Switzerland
{firstname.lastname}@epfl.ch

Abstract

The main goal of modeling human conversation is to create agents which can interact with people in both open-ended and goal-oriented scenarios. End-to-end trained neural dialog systems are an important line of research for such generalized dialog models as they do not resort to any situation-specific handcrafting of rules. However, incorporating personalization into such systems is a largely unexplored topic as there are no existing corpora to facilitate such work. In this paper, we present a new dataset of goal-oriented dialogs which are influenced by speaker profiles attached to them. We analyze the shortcomings of an existing end-to-end dialog system based on Memory Networks and propose modifications to the architecture which enable personalization. We also investigate personalization in dialog as a multi-task learning problem, and show that a single model which shares features among various profiles outperforms separate models for each profile.

1 Introduction

The recent advances in memory and attention mechanisms for neural networks architectures have led to remarkable progress in machine translation [Johnson et al., 2016], question answering [Graves et al., 2016] and other language understanding tasks which require an element of logical reasoning. The main motivation for building neural network based systems over traditional systems for such tasks is that they do not require any feature engineering or domain-specific handcrafting of rules [Vinyals and Le, 2015]. Conversation modeling is one such domain where end-to-end trained systems have matched or surpassed traditional dialog systems in both open-ended [Dodge et al., 2015] and goal-oriented applications [Bordes and Weston, 2016].

An important yet unexplored aspect of dialog systems is the ability to personalize the bot's responses based on the profile or attributes of who it is interacting with. Personalization is key to creating conversational agents that are truly smart and can integrate seamlessly into the lives of human beings. However, there are no open datasets which allow researchers to train end-to-end dialog systems where each conversation is influenced by a speaker's profile [Serban et al., 2015].

With the ultimate aim of creating such a dataset, this paper aims to be an extension of the bAbI dialog dataset introduced by Bordes and Weston [2016]. Set in the domain of restaurant reservation, their synthetically generated dataset breaks down a conversation into several tasks to test some crucial capabilities that dialog systems should have. Taken together, the tasks can be used as a framework for the analysis of end-to-end dialog systems in a goal-oriented setting. Given a knowledge base (KB) of restaurants and their properties (location, type of cuisine, etc.), the aim of the dialog is to book a restaurant for the user.

In this paper, we propose extensions to the first five tasks of the existing dataset. In addition to the goal of the original task, the dialog system must leverage a user's profile information to alter speech

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style and personalize reasoning over the KB. The end-goal is to make a restaurant reservation that is personalized to the user's attributes (dietary preference, favorite food items, etc.).

The synthetic nature of the bAbI dialog dataset and by extension, our work, makes it easy to construct a perfect handcrafted dialog system. Hence, the goal here is not to improve the state of the art in this domain, but to analyze existing end-to-end goal-oriented dialog systems and to model personalization in such frameworks. Section 3 presents our modifications to the original dataset and Section 4 describes the various models that are benchmarked on our tasks in Section 5. In Section 6, we analyze the task of conducting goal-oriented dialog as a multi-task learning problem, and show that a single model which shares features among various profiles outperforms separate models for each profile.

Our dataset is accessible through Github along with our experimental code and trained models.[1]

2 Related Work

This work builds upon the bAbI dialog dataset described in Bordes and Weston [2016], which is aimed at testing end-to-end dialog systems in the goal-oriented domain of restaurant reservations. In a similar investigation, Li et al. [2016] encoded speaker persona into SEQ2SEQ dialog models [Vinyals and Le, 2015]. Their focus is on open-ended dialog and they build an embedding of a persona in a vector space based on conversation history of the speaker. However, our work is focused on composing the user's profile during conversation from explicitly provided attributes.

3 Dataset Creation

We build upon the first five synthetically generated bAbI dialog tasks (T1-T5) where the goal is to book a table at a restaurant. The conversations are generated by a simulator (in the format shown in Figure 1) based on a KB containing multiple restaurants and their properties. Each restaurant is defined by a type of cuisine, a location, a price range, a party size and a rating (from 1 to 8). Making an API call to the KB returns a list of facts related to all the restaurants that satisfy the parameters.

Conversations are generated using natural language patterns after randomly selecting each of the four required fields: location, cuisine, price range and party size. There are 43 patterns for the user and 15 for the bot (the user can say something in up to 4 different ways, while the bot only has one).

We make further additions to the KB and augment the bot utterance patterns for the creation of our tasks. To fit in with the synthetic nature of the bAbI dialog tasks, the personalization of the bot's responses are handcrafted to be extremely simplistic in comparison to real life situations.

3.1 Personalized Goal-Oriented Dialog Tasks

In addition to fulfilling the original goals of the bAbI dialog tasks, the modified tasks also require the dialog system to personalize its speech style and reasoning over KB facts based on the user's profile, which is composed of various fixed attributes. The attributes (gender, age, dietary preference and favorite food item) are provided before the first turn of each dialog.

Personalization Task 1: Issuing API calls Users define a query containing from 0 to 4 of the required fields (sampled uniformly). The bot must ask questions to fill the missing fields and then generate the proper API call.

Personalization Task 2: Updating API calls Starting by issuing an API call as in Task 1, users then ask to update their requests between 1 and 4 times (sampled uniformly). The fields to update are selected randomly and the bot must then issue the updated API call.

Personalization Task 3: Displaying Options Given a user request, the KB is queried by the corresponding API call and the resulting facts are added to the dialog history. The bot must sort the restaurants in the facts using simple heuristics (described in Section 3.3) based on the user's attributes and propose a restaurant to the users until they accept. Users accept a suggestion 25% of the time or always if it is the last remaining one.

Personalization Task 4: Providing extra information Given a user request for a randomly sampled restaurant, all KB facts related to the restaurant are added to the history and the dialog is

[1] `https://github.com/chaitjo/personalized-dialog`

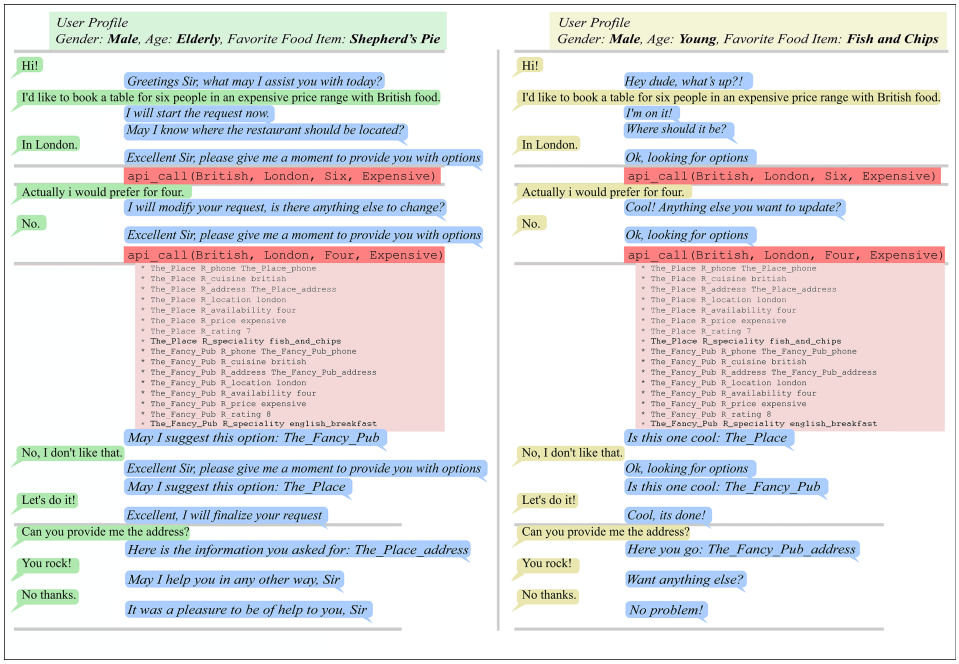


Figure 1: **Personalized Restaurant Reservation System.** The user (in green or yellow) conducts a dialog with the bot (in blue) to reserve a table at a restaurant. At each turn, a model has access to the user's profile attributes, the conversation history and the outputs from the API call (in light red) and must predict the next bot utterance or API call (in dark red). The horizontal lines between dialog groups signify the separate tasks that are described in the following sections. (Illustration adapted from Figure 1, Bordes and Weston [2016].)

conducted as if the user has agreed to book a table there. The user then asks for the directions to the restaurant, its contact information or both (with probabilities 25%, 25% and 50% respectively). The bot must learn to retrieve the correct KB facts from history, tailored for the user.

Personalization Task 5: Conducting full dialogs Conversations generated for task 5 combine all the aspects of tasks 1-4 into full dialogs.

3.2 User Profiles and Speech Style Changes

The first aspect of personalization incorporated into all 5 tasks was the change in the style of the language used by the bot based on the user's gender (male or female) and age (young, middle-aged or elderly). For each of the 15 bot utterance patterns in the original tasks, we created 6 new patterns for each possible (age, gender) profile permutation.[2] Hence, we increased the size of each task by 6 folds. Each of these patterns, while conveying the same information to the user, differed in tone, formality and word usage. Figure 1 illustrates two versions of the same dialog for a (male, elderly) user and a (male, young) user.

While creating the utterance patterns, importance was given to maintaining a consistent vocabulary for each of the 6 profiles. The levels of formality and precision of the words and language used by the bot increased with age. At the same time, word choices overlapped between the same gender and age group. For example, the pattern for a (male, young) user is similar in formality and tone to the pattern for a (female, young) user and shares certain key words with both (female, young) and (male, middle-aged) user patterns. It is comparatively unrelated to the patterns of a (female, middle-aged) or (female, elderly) user. By creating such relationships between the profiles instead of having 6 completely distinct patterns, we wanted to test whether dialog models could learn to form associations between concepts (such as formality, precision and word usage) and attributes (such as gender and age).

[2]Table of all utterance pattern changes: `https://github.com/chaitjo/personalized-dialog/blob/master/speechstyle.md`.

3.3 KB Updates and Personalized Reasoning

To personalize the order in which restaurants are recommended by the bot in task 3, we added 2 new attributes to the user's profile: dietary preference (vegetarian or non-vegetarian) and favorite food item (randomly sampled from a list of dishes associated with the cuisine in the dialog). We created a duplicate for each restaurant in the KB, with an additional attribute for type of restaurant (vegetarian or non-vegetarian) to differentiate the otherwise same copies. For every restaurant in the modified KB, we also added the speciality attribute (randomly sampled from a list of dishes associated with the restaurant's cuisine). When modifying each dialog in the original task, instead of proposing the restaurants solely based on their rating (as in the bAbI dialog tasks), we used a score calculated as: rating (out of 8) + 8 (if restaurant type matches user's dietary preference) + 2.5 (if restaurant speciality matches user's favorite food item). With such a metric, all vegetarian restaurants will be scored 8 points higher than non-vegetarian restaurants for a vegetarian user and will always be proposed before the bot suggests any non-vegetarian ones (in descending order of score). By awarding an extra 2.5 points to reflect a user's preference for a favorite food item, a lower rated restaurant specializing in the item will be proposed before a higher rated (by at most 2 points) restaurant specializing in something else. This tests a model's ability to perform true/false reasoning based on the user's profile and to implicitly rank restaurants depending on more than one condition.

Our modification to task 4 requires the bot to retrieve a combination of KB facts related to a restaurant based on certain attributes of the user and the restaurant itself. In addition to the phone number and address, we added 3 new attributes (social media links, parking information and public transport information) to the KB entries for every restaurant. In each modified dialog, when a user asks for the contact information of the restaurant, the bot must return the restaurant's social media link if the user is young, or the phone number if the user is middle-aged or elderly. Similarly, when a user asks for the directions to the restaurant, the bot must return the address and the public transport information if the restaurant is cheap, or the address and the parking information if it is in the moderate or expensive price range. This tests a model's ability to personalize KB fact retrieval based on an attribute in the user's profile (age) or a choice made by the user during the dialog (the restaurant's price range).

3.4 Updated Dataset

We generated and structured the dataset in the same way as the original bAbI dialog dataset: for each task, we provided training, validation and test set dialogs generated using half of the modified KB. We also generated another test set from the remaining KB containing new entities (restaurants, cuisine types, etc.) unseen in any training dialog, called Out-Of-Vocabulary (OOV) test set. During training, the model has access to training examples and the KB. Models can be evaluated on both test sets, plain and OOV, on their ability to rank the correct bot utterance at each turn of the dialog from a list of all possible candidates.

The statistics of the datasets for each task are given in Table 1, along with a comparison to the original bAbI dialog tasks. We provide two variations for each task: a full set with all generated dialogs and a small set with only 1000 dialogs each for training, validation and testing to create realistic learning conditions.

Table 1: **Dataset statistics.** For rows 4 to 6, the first number is the size of the full set and the number in parenthesis is the size of the small set. (*) PT1-PT5 and all 5 bAbI dialog tasks have two test sets of the same size, one using the same KB entities as the training set and the other using out-of-vocabulary words.

Task	PT1	PT2	PT3	PT4	PT5	bAbI dialog tasks (T1-T5)
Vocabulary size			14819			3747
Candidate set size			43863			4212
Training dialogs	6000 (1000)	6000 (1000)	12000 (1000)	6000 (1000)	12000 (1000)	1000 each
Validation dialogs	6000 (1000)	6000 (1000)	12000 (1000)	6000 (1000)	12000 (1000)	1000 each
Test dialogs	6000* (1000*)	6000* (1000*)	12000* (1000*)	6000* (1000*)	12000* (1000*)	1000* each

4 Models

Following Bordes and Weston [2016], we provide baselines on the modified dataset by evaluating the end-to-end Memory network model and propose modifications to the Memory Network architecture that facilitate personalized reasoning over KBs.

4.1 Memory Networks

Memory Networks [Weston et al., 2014] are a recent class of models that have proven successful for a variety of tasks such as question answering [Sukhbaatar et al., 2015] and conducting dialog [Dodge et al., 2015]. For dialogs, the entire conversation history is stored in the memory component of the model. It can be iteratively read from to perform reasoning and select the best possible responses based on the context. The model's attention to various entries in memory at each turn of conversation can be visualized over multiple iterations (called hops).

The user profile information is stored in the memory of the model as if it were the first turn of the conversation history spoken by the user, i.e. the model builds an embedding of the profile by combining the values of the embeddings of each attribute in the profile. Our goal is to analyze the capabilities of the existing Memory Network model to leverage profile information when conducting dialog.

4.2 Split Memory Architecture

The personalized reasoning aspect of Tasks 3 and 4 require the bot to combine information from the user's attributes as well as the conversation history (i.e. KB facts and dialogs). A single memory architecture is not be able to focus sufficiently on the various attributes and dialog turns used to formulate a personalized response.

We propose dividing the memory of the model into two halves: profile attributes and conversation history. The various attributes are added as separate entries in the profile memory before the dialog starts. At the same time, each dialog turn is added to the conversation memory as the dialog progresses. The mechanism through which the model reads from and reasons over each memory remains the same. The outputs from both memories are summed element-wise to get the final response for each conversation turn. A visualization of the model is shown in Figure 2.

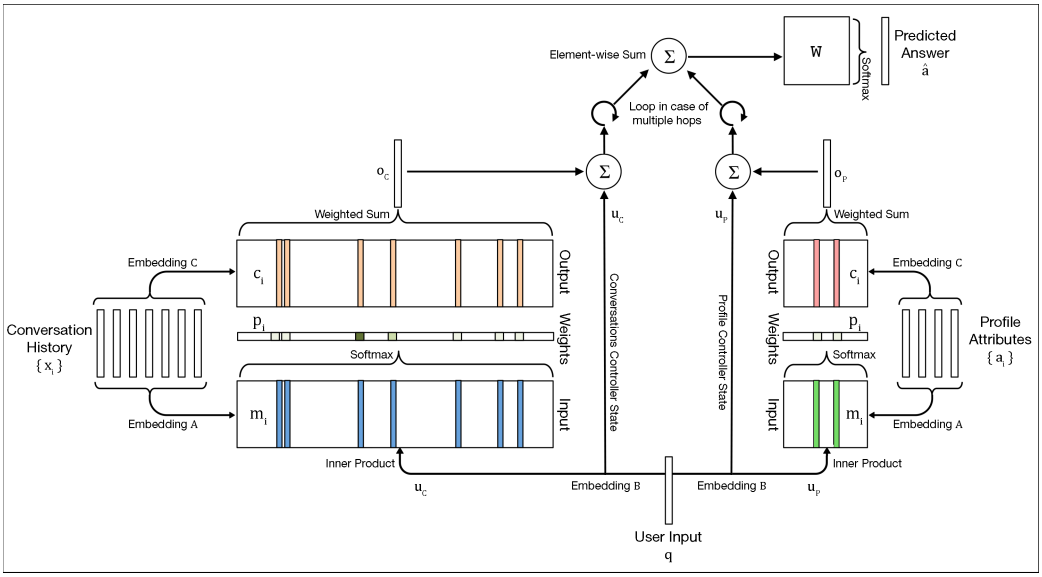


Figure 2: **Split memory architecture for Memory Networks.** Profile attributes and conversation history are modeled in two separate memories. The outputs from both memories are summed to get the final response.

5 Experiments

We report per-response accuracy (i.e. the percentage of responses in which the correct candidate is chosen out of all possible ones) across the models and tasks in Table 2. The rows show tasks PT1-PT5 and columns 2-3 give the accuracy for each of the models. The hyperparameters for the models were optimized on the validation sets. Memory Networks completed PT1 and PT2 (issuing and updating API calls) with a very high degree of accuracy. This indicates that the model is able to implicitly track dialog state and personalize the bot's utterance based on the user's profile.

Table 2: **Test results across models and tasks.** The first number is the accuracy on the full set of dialogs for each task and the number in parenthesis is the accuracy on the small set (with 1000 dialogs).

Task	Memory Networks	
	Standard	Split memory
PT1: Issuing API calls	99.83 (98.87)	85.66 (82.44)
PT2: Updating API calls	99.99 (99.93)	93.42 (91.27)
PT3: Displaying options	58.94 (58.71)	68.60 (68.56)
PT4: Providing information	57.17 (57.17)	57.17 (57.11)
PT5: Full dialog	85.10 (77.74)	87.28 (78.1)

Results on PT3-PT5 suggest that Memory Networks were unable to use KB facts and profile information in conversation reliably. Memory Networks with the *split memory* feature significantly outperformed the standard architecture for PT3, which lead to better performance on the full dialog task PT5. Visualization and analysis of the model's attention weights in Tables 5 and 6 show that standard Memory Networks fail to interpret knowledge about entities and link it to the user's attributes, while splitting memory helps to combine profile information with the correct KB facts when sorting restaurants (in PT3) or providing additional information about them (in PT4).

Despite extensive hyperparameter tuning, the *split memory* model proved hard to train for the PT1 and PT2. Training converged at significantly lower accuracy than the standard model, resulting in lower test accuracy and suggesting that a simpler model is more suitable for tasks which do not require compositional reasoning over various entries in the memory.

Although no model was able to solve PT3 and PT4 sufficiently, results with the *split memory* architecture for Memory Networks are encouraging. We believe that such a modification is not specific to Memory Networks and can be applied to any generic dialog model which stores and reasons over conversation turns.

6 Multi-task Learning

We analyzed the Memory Network architecture in a multi-task learning scenario for conducting full dialog. We trained individual profile-specific models for each of the 6 profile permutations for speech style changes, and compared their performance to a single multi-profile model. Each of the profile-specific models were trained on 1000 full dialogs between the bot and a user with the corresponding age and gender combination (6000 dialogs in total). The multi-profile model was trained on the same 6000 full dialogs, containing all 6 user profiles. For each profile, we report per-response accuracies for both the profile-specific and multi-profile model on 1000 test dialogs (with users having the same profile) in Table 3. The multi-profile model significantly outperformed each profile-specific model, indicating that training a single model on dialogs with multiple profiles which share logic and vocabulary is an effective learning strategy.

Table 3: **Test results for multi-task learning scenarios.**

Profile	Profile-specific model	Multi-profile model
male, young	80.38	85.04
female, young	80.15	84.91
male, middle-aged	80.29	84.71
female, middle-aged	80.21	85.14
male, elderly	80.57	85.46
female, elderly	80.41	84.95

Table 4: **Attention weights of multi-profile model versus profile-specific model.**

Time	Locutor	Dialog History	Multi-profile model	Profile-specific model
1		female middle-aged non-veg pizza	0.044	0
2	User	good morning	0.397	0
3	Bot	hello maam how can i help you	0.005	0.012
4	User	can you make a restaurant reservation for two people in bombay with italian cuisine	0.533	0.987
5	Bot	give me a second for processing the reservation	0	0
User input		<SILENCE>		
Correct answer		which price range are you looking for		
Predicted answer		which price range are you looking for (for both models)		

We attribute the increase in accuracy to learning shared features among the 6 user profiles, which can be illustrated by comparing the attention weights for the two strategies in Table 4. Given the same user profile, the multi-profile model attends to the user's attributes, greeting and incomplete inquiry to modify its speech style and ask for the missing field, while the profile-specific model has a narrow focus on the inquiry.

7 Conclusion

This paper aims to bridge a gap in research on neural conversational agents by introducing a new dataset of goal-oriented dialogs with user profiles associated with each dialog. Crucial aspects of goal-oriented conversation have been split into various synthetically generated tasks to evaluate the strengths and weaknesses of models in a systematic way before applying them on real data. Despite the language and scenarios of the tasks being artificial, we believe that building mechanisms that can solve them is a reasonable starting point towards the development of sophisticated personalized dialog systems in domains such as restaurant reservation, customer care or digital assistants.

Additionally, we demonstrated how to use our tasks to break down a dialog system based on Memory Networks. The model was unable to perform compositional reasoning or personalization to solve the tasks, indicating that further work needs to be done on learning methods for these aspects. Towards this goal, we proposed a *split memory* architecture for end-to-end trained dialog agents that lead to an improvement in overall performance.

Finally, we demonstrated the advantages of formulating personalization in dialog as a multi-task learning problem as opposed to training separate models for various user profile. We showed that a jointly trained model significantly outperformed profile-specific models and is able to leverage shared features and relationships between the various conversation styles.

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Appendix

Table 5: **Attention Weights for PT3 (Displaying options).** The model should ideally be focusing on the user's profile and the ratings, types and specialities of the various restaurants in the KB facts. However, the standard Memory Network does not show any meaningful attention to the user profile or KB facts used for implicit ranking, indicating that the model is insufficient at reasoning over the KB. The *split memory* model pays special attention to the restaurants that have already been suggested. It also attends to the user's profile in a meaningful way: it iterates over the dietary preferences and favorite food item to perform reasoning, and then over the gender and age to modify speech style. The predicted response is more meaningful than the standard Memory Network, although it is unable to distinguish between the embeddings of various restaurants. (We have only shown important utterances in the table.)

			Standard Model			Split Memory Model		
		Profile	**Hop #1**	**Hop #2**	**Hop #3**	**Hop #1**	**Hop #2**	**Hop #3**
		female				0.011	0.571	0
		young	0.0001	0	0	0.017	0.423	0
		non-veg				0.442	0.006	0.999
		pizza				0.53	0	0
Time	**Locutor**	**Dialog History**						
19	User	resto_rome_moderate_italian_8stars_1 R_rating 8	0.0001	0	0	0.0001	0	0
20	User	resto_rome_moderate_italian_8stars_1 R_type veg	0	0	0	0	0	0
21	User	resto_rome_moderate_italian_8stars_1 R_speciality pizza	0	0	0	0.0002	0	0
23	User	resto_rome_moderate_italian_8stars_1 R_parking resto_rome_moderate_italian_8stars_1_parking	0	0	0	0.0127	0	0
29	User	resto_rome_moderate_italian_3stars_1 R_location Rome	0.879	0.999	0	0	0	0
31	User	resto_rome_moderate_italian_3stars_1 R_rating 3	0	0	0	0	0	0
33	User	resto_rome_moderate_italian_3stars_1 R_type veg	0.0001	0	0	0	0	0
43	User	resto_rome_moderate_italian_2stars_2 R_rating 2	0	0	0	0	0	0
44	User	resto_rome_moderate_italian_2stars_2 R_type non-veg	0.0001	0	0	0	0	0
45	User	resto_rome_moderate_italian_2stars_2 R_speciality pasta	0	0	0	0	0	0
51	User	resto_rome_moderate_italian_2stars_1 R_address resto_rome_moderate_italian_2stars_1_address	0	0	0	0.0184	0	0
73	User	hi	0	0	0	0.0002	0	0
74	Bot	hey girl how is it going	0	0	0	0.0006	0	0
75	User	may i have a table in rome with italian food in a moderate price range	0	0	0	0.0022	0	0
76	Bot	be right back with your reservation	0	0	0.999	0	0	0
80	Bot	sure finding some options	0	0	0	0.0002	0.0002	0.999
81	User	<SILENCE>	0	0	0	0.0008	0	0
82	Bot	how about this one: resto_rome_moderate_italian_8stars_2	0	0	0	0.3872	0.003	0
83	User	no i don't like that	0	0	0	0	0	0
84	Bot	sure finding something else	0	0	0	0.0001	0	0
85	User	<SILENCE>	0.0003	0	0	0.001	0	0
86	Bot	how about this one: resto_rome_moderate_italian_3stars_2	0	0	0	0.4834	0.9968	0
87	User	no i don't like that	0	0	0	0	0	0
88	Bot	sure finding something else	0	0	0	0.0001	0	0
User Input		<SILENCE>						
Answer		**(Correct)** how about this one: resto_rome_moderate_italian_8stars_1	what food are you looking for			how about this one: resto_paris_cheap _italian_2star_1		

Table 6: **Attention Weights for PT4 (Providing information).** The standard model directs its attention to all the KB facts that it may need to provide but does not focus on the user profile sufficiently. Instead, it also attends to its own final utterance before the turn, which may have helped it judge the user's gender and age instead of using the profile. It correctly predicts that it has to display the social media information instead of the phone number for the young user, but provides the information for the wrong restaurant. The *split memory* model is able to single out the important user attribute (age) needed to reason over the correctly identified KB facts. However, it still mixes up embeddings of similar KB entities despite the stronger signals compared to the standard Memory Network.

			Standard Model			Split Memory Model		
		Profile	**Hop #1**	**Hop #2**	**Hop #3**	**Hop #1**	**Hop #2**	**Hop #3**
		male	0.003	0	0	0.431	0.091	0.002
		young				0.569	0.909	0.998
Time	**Locutor**	**Dialog History**						
1	User	resto_madrid_cheap_indian_8stars_1 R_phone resto_madrid_cheap_indian_8stars_1_phone	0.379	0.311	0.001	0.001	0	0
2	User	resto_madrid_cheap_indian_8stars_1 R_cuisine indian	0.001	0	0	0	0	0
3	User	resto_madrid_cheap_indian_8stars_1 R_address resto_madrid_cheap_indian_8stars_1_address	0.021	0.005	0	0	0	0
4	User	resto_madrid_cheap_indian_8stars_1 R_location madrid	0.003	0.015	0.342	0	0	0
5	User	resto_madrid_cheap_indian_8stars_1 R_number eight	0.002	0	0	0	0	0
6	User	resto_madrid_cheap_indian_8stars_1 R_price cheap	0.002	0.001	0	0	0	0
7	User	resto_madrid_cheap_indian_8stars_1 R_rating 8	0.001	0	0	0	0	0
8	User	resto_madrid_cheap_indian_8stars_1 R_type veg	0.001	0	0	0	0	0
9	User	resto_madrid_cheap_indian_8stars_1 R_speciality biryani	0.002	0	0	0	0	0
10	User	resto_madrid_cheap_indian_8stars_1 R_social_media resto_madrid_cheap_indian_8stars_1_social_media	0.084	0.069	0.001	0.21	0	0
11	User	resto_madrid_cheap_indian_8stars_1 R_parking resto_madrid_cheap_indian_8stars_1_parking	0.354	0.222	0	0.385	0	0
12	User	resto_madrid_cheap_indian_8stars_1 R_public_transport resto_madrid_cheap_indian_8stars_1_public_transport	0.019	0.001	0	0.066	0	0
13	User	hello	0.001	0	0	0	0	0
14	Bot	hey dude what is up	0	0	0.004	0.309	0.999	0.999
15	User	can you make a restaurant reservation at resto_madrid_cheap_indian_8stars_1	0.005	0	0	0.021	0	0
16	Bot	cool its done	0.116	0.37	0.652	0.005	0	0
User Input		may i have the contact details of the restaurant						
Answer		**(Correct)** here you go resto_madrid_cheap_indian_8stars_1_social_media	here you go resto_rome_cheap_indian _7star_2_social_media			here you go resto_paris_cheap_indian _2star_2_social_media		